

IEOR E4004

Optimization Models and Methods

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Outline

- Introduction to Solvers | Gurobi Example
- Process of Optimization in Real Life
- Introduction to Large Scale Data Driven Optimization
 - A Shipment Problem

Introduction to Solvers

Gurobi

Gurobi | Vaccine Allocation Problem



Problem Statement

A hospital has received a batch of 5000 COVID-19 vaccines. The hospital has 4 vaccine stations - Station A, Station B, Station C, and Station D. Each station has a priority based on the population it serves (e.g., elderly, children, chronically ill, disabled). The higher the priority value, the more vulnerable the population it serves. Additionally, each station has a storage capacity, and it is mandatory that every station receives at least 500 vaccines.

Gurobi | Vaccine Allocation Problem



Table 1: Station priorities and storage capacities.

Station	Priority	Storage Capacity (Vaccines)	Demand (Vaccines)
A	0.5	2300	2000
B	0.7	1700	2400
C	1.0	1200	1600
D	0.8	900	1000

Please formulate this problem as a linear program and find the optimal vaccine allocation strategy for this hospital using Gurobi.

Gurobi | Vaccine Allocation Problem



Mathematical Model

Decision variables:

X_i : number of vaccines allocated to station i , $\forall i \in \{A, B, C, D\}$

Parameters:

P_i : priority value of station i , $\forall i \in \{A, B, C, D\}$

S_i : storage capacity at station i , $\forall i \in \{A, B, C, D\}$

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Objective Function:

This function is designed to maximize the total “benefit” achieved from distributing the vaccines.

The benefit is calculated as the product of the number of vaccines allocated to each station and the priority of that station.

$$\text{Maximize} \quad \sum_{i \in \{A, B, C, D\}} P_i \cdot X_i$$

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Constraints:

1. Storage space constraints: This constraint ensures that the number of vaccines allocated to each station does not exceed the station's storage capacity. S_i is the storage capacity of station i .

$$X_i \leq S_i, \quad \forall i \in \{A, B, C, D\}$$

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2. Use all vaccines: This constraint guarantees that all available vaccines are allocated. The total number of vaccines available is 5000, so the sum of the vaccines allocated to all stations must equal this number.

$$\sum_{i \in \{A, B, C, D\}} X_i = 5000$$

Gurobi | Vaccine Allocation Problem



3. Minimum supply constraints: This constraint ensures that every station receives at least 500 vaccines. The decision variable X_i represents the number of vaccines allocated to each station, and i is the index of the stations.

Note: No need for non-negativity constraints.

$$X_i \geq 500, \quad \forall i \in \{A, B, C, D\}$$

The Process of Optimization in the Real World

Step 1: Problem Definition



- Identify the problem faced by the decision-maker
- Clearly define the **objective** of the analysis
- Understand the **system** that needs to be studied
- Set the scope for modeling the real-world issue



Step 2: Data Collection

- Gather data that is relevant to the decision model
- May involve:
 - Observing the system firsthand
 - Using existing recorded data
- Ensures the model is grounded in factual and usable information



Step 3: Model Formulation

- Translate the real-world problem into a mathematical model
- Use **mathematical language** to express the decision problem
- Simplify complex operations into manageable components
- Focus on clarity and accuracy in representing the problem

Step 4: Model Solution



- Solve the model to find an **optimal or near-optimal** solution
- Typically done using **numerical algorithms**, not algebra
- Aim is to support decision-making with precise, calculated outcomes

Step 5: Verification and Validation



Verification: Check that the model aligns with its assumptions and structure

Validation: Ensure the model produces accurate, useful results

Methods used include:

- Testing across various feasible solutions
- Statistical or simulation-based validation techniques

Step 6: Implementation



- Apply the solution in practice
- Adapt results if necessary (e.g., convert fractional to integer values)
- Provide insights to aid decision-makers
- Develop systems for repeated use with new data for future decisions

Introduction to Large Scale Data Driven Optimization

A Shipment Example

Context:

- A global seller is launching 15 new products (Product1 to Product15).
- Customers worldwide have placed pre-orders.
- The company must fulfill these orders **cost-effectively** using 10 warehouses (Warehouse1 to Warehouse10).

Goal:

- Determine the **optimal shipping plan** that minimizes shipping costs.
- Costs are proportional to the **distance** from warehouse to customer.



Understanding the Data

3 Key Inputs:

1. **Pre-Order Matrix** – (859 customers × 15 products)
→ How many units each customer ordered.
2. **Stock Matrix** – (10 warehouses × 15 products)
→ Available stock of each product at each warehouse.
3. **Distance Matrix** – (10 warehouses × 859 customers)
→ Distance between every warehouse and every customer.

Assumption:

Shipping cost $\propto$ Distance

Mathematical Formulation - Linear Programming

Sets:

- Warehouses $W = \{1, \dots, 10\}$
- Customers $C = \{1, \dots, 859\}$
- Products $P = \{1, \dots, 15\}$

Decision Variable:

x_{pwc} : Quantity of product p shipped from warehouse w to customer c

Objective:

$$\text{Minimize } \sum_{p \in P} \sum_{w \in W} \sum_{c \in C} d_{wc} \cdot x_{pwc}$$

Constraints

1. Stock Constraint:

Total units shipped from a warehouse $\leq$ its stock

$$\sum_{c \in C} x_{pwc} \leq s_{pw}$$

1. Pre-Order Fulfillment:

Total units received by each customer = their order

$$\sum_{w \in W} x_{pwc} = q_{pc}$$

1. Non-Negativity:

$$x_{pwc} \geq 0$$

Optional: Add integrality constraint if integer shipping units are required.

Gurobi Implementation Highlights

- Load and preprocess data from Excel.
- Define sets, parameters, and decision variables in Gurobi.
- Add the objective function and constraints.
- Solve the LP (or ILP).
- Output optimal shipment plan (values of x_{pwc}).

Takeaway:

The model successfully finds the **lowest-cost shipping plan** that meets customer demand and respects warehouse stock limits.

Product Distribution from Each Warehouse

